

Assignment: 1 NLP

Q.1 Explain the different phases of NLP with a neat diagram.

- Natural language processing (NLP) is a field of artificial intelligence that enables computers to understand, interpret and generate human language. NLP works in multiple phases to convert raw text into meaningful information.

Phases of NLP:

1. Lexical Analysis

- First phase of NLP
- Converts input text into tokens (words, number, punctuation).
- Removes unnecessary symbols.
- Identifies base word forms.

2. Syntax analysis (parsing)

- Checks grammatical structure of the sentence.
- Determines subjects, verb, object relationship.
- Uses grammar rules.
- Builds parse tree

3. Semantic Analysis

- Determine the meaning of word and sentence.
- Resolves ambiguity.
- Assign correct meaning based on context.
- Creates semantic representation.

4. Discourse integration

- Connect meaning across multiple sentences

- Maintains Context
- Resolves pronouns (he, she, it).
- Ensures logical flow.

5. Pragmatic analysis.

- Understand intended meaning
- Uses real-world knowledge.
- Consider speaker intention
- Interprets implied meaning

Diagram :

Input Text

Lexical Analysis

Syntax Analysis

Semantic Analysis

Discourse Integration

Pragmatic Analysis

Final Meaning / Output

Describe the process of lexical analysis in NLP.
What are tokenization and stemming?

Lexical Analysis is the first step in NLP where raw text is converted into meaningful units.

- process of lexical Analysis
1. Input Sentence is received
 2. Sentence is divided into tokens
 3. Punctuation marks are removed
 4. Stop words may be removed
 5. Words are converted to lower case
 6. Root forms of words are identified

★ Tokenization : is the process of breaking text into smaller units called tokens.
 Example: Sentence : Code.

NLP is very interesting.

After Tokenization :

["NLP", "is", "very", "interesting"]

TYPES : Word Tokenization
 . Sentence Tokenization

Stemming :

reduces a word to its root form by removing Suffixes.

Word	Stem
Playing	Play
Running	Run
Studies	Studi-

Stemming is simple & fast but may not always give correct root word.

3. Demonstrate how Named Entity Recognition (NER) can be used to extract entities from a Paragraph.

- Named Entity Recognition (NER) is an NLP technique used to identify and classify important entities in text such as:

- Person
- Organization
- Location
- Date
- Time
- Money
- Percentage

Example Paragraph:

"Anirudh Behara" is studying at Anirudh University Mumbai. He lives in Mumbai and will join TCS in July 2026.

Entities Extracted using NER

Entity Type

Extracted Entity

Person

Anirudh Behara

Organization

Anirudh University Mumbai

Location

Mumbai

Organization

TCS

Date

July, 2026

NER is widely used in:

- Chatbot
- Resume Screening
- News Analysis
- Search engines

NER Working Diagram:

Input paragraph

Tokenization

Model Processing

Entity Classification

Structured output (person, location, org, date)

4. Analyze the difference between Syntax analysis and Semantic analysis in NLP.

Syntax analysis and Semantic analysis are two important phases of NLP but they focus on different aspects.

Syntax analysis (parsing)

Checks grammatical correctness.

Focuses on structure

Builds parse tree

uses grammar rules

Example: "She go to school" → incorrect grammar

Semantic analysis: Focuses on meaning

Interprets word sense

Resolves ambiguity

checks logical correctness

Example: "Colours green"

→ Grammatically correct but meaningless

Comparison Table:

	Syntax Analysis	Semantic Analysis
i.	Structure based	Meaning based
ii.	Grammar rule used	Context & knowledge used
iii.	Creates parse tree	Creates meaning representation
iv.	Detects grammatical error	Detects meaning error
v.	Earlier phase	After syntax analysis

Diagram:

Sentence

Syntax Analysis → checks grammar

Semantic Analysis → checks meaning

5. Examine the role of discourse analysis in understanding multi-sentence text.

20 Discourse analysis studies how multiple sentences connect and form meaningful communication. It ensures text coherence and context understanding.

Role of discourse analysis

1. Resolving pronouns

Example:

"Ravi bought a car. He loves it."

• He → Ravi

• it → Car

2. Maintaining Context

understanding topic continuity

3. Logical flow

Ensure ideas are connected.

4. Identifying Relationship
Cause-effect, Contrast, Sequence.

5. Removing Ambiguity.
Clarifies reference across sentence.

Example:

Text:

"Anita went to the market. She bought apples."

They were fresh."

DisCours Analysis Identifies:

• She → Anita

• They → Apples

Diagram of Discours analysis

Sentence 1 → Meaning 1